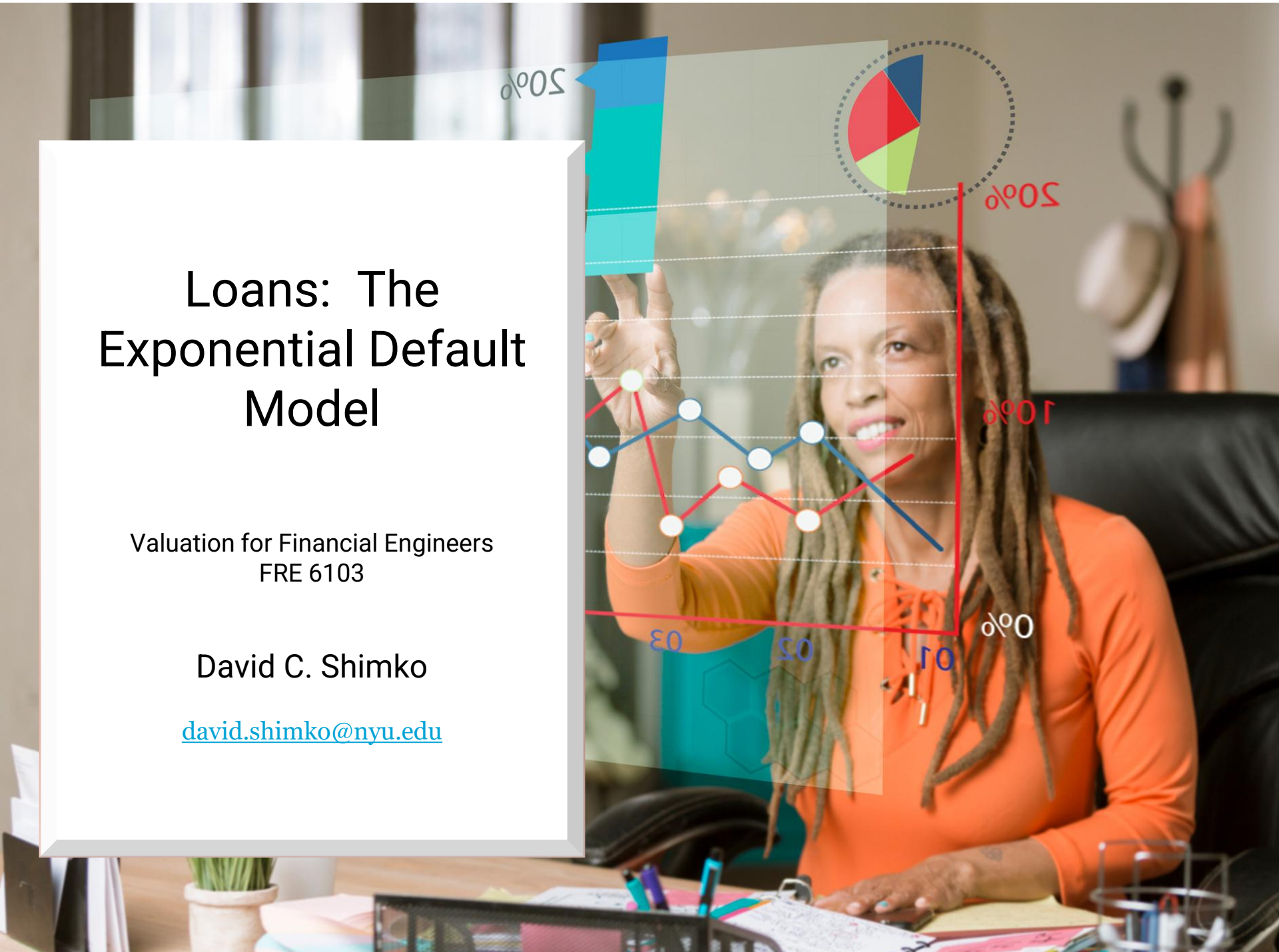
Loans: The Exponential Default Model

Valuation for Financial Engineers
FRE 6103

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Application: Consumer Loans

- Let's say you need to borrow $5000
- Your bank will allow you to borrow for 24 months at an annual interest rate of 6%
- The loan will be fully amortized (nothing due at the end)
- You have a lot of questions:
  - What are the monthly payments?
  - What is my total cost of borrowing $5000?
  - What do I owe if I pay off early?



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Step 1: Payment & first month amortization

- Which of the following best represents the cash flow pattern?
  - Single payment, annuity, perpetuity, growing annuity, or combination?
- How can we calculate the payment?
  - In Excel, $221.60=PMT(0.06/12,24,-5000)
- What happens in the first month?
  - At the end of the first month ($5000 \times 0.005 = \$25$) is due in interest
  - The payment of $221.60 is used to pay the interest first
  - The remainder (221.60 – 25 =) $196.60 is used to reduce the principal
  - The ending balance of the first month becomes the beginning balance of the second month

AMORTIZATION TABLE					
Month	Begin	Payment	Interest	Prin Redox	End
1	$5,000.00	221.60	25.00	196.60	4,803.40
2	4,803.40				

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Step 2: Complete the amortization table

- Month 1 is simple to copy to the other months if you use absolute and relative addressing in Excel
- If your table is completed correctly, the final balance after month 24 is *exactly zero*
- The "End" column shows the balance of the loan if it is prepaid in any month
  - e.g. If paid off in month 15, balance is $1945.47
- The ending column always equals the present value of the remaining payments
  - e.g. At end of month 15, 9 payments remain, so
  - 1945.47 = -PV(0.005,9,221.60)
- Do this in Excel to get the exact values!

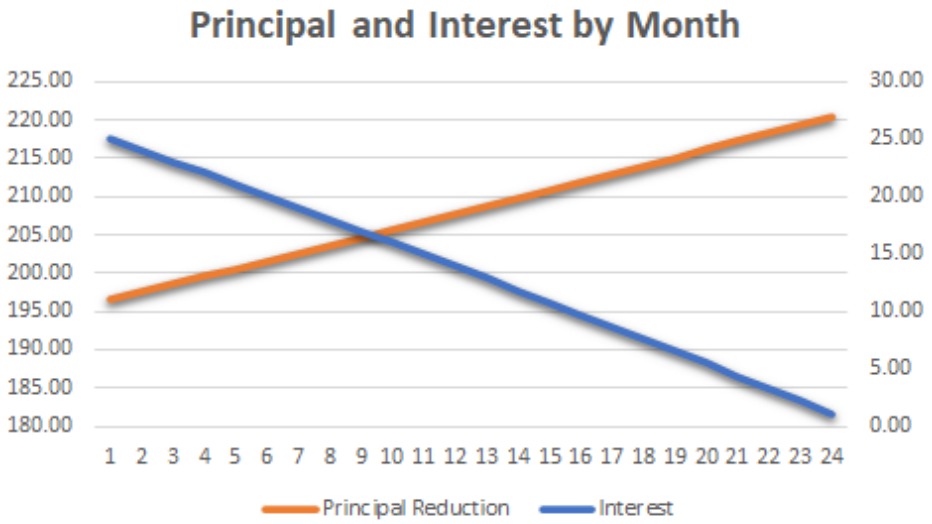
AMORTIZATION TABLE

Month	Begin	Payment	Interest	Prin Redox	End
1	$5,000.00	221.60	25.00	196.60	4,803.40
2	4,803.40	221.60	24.02	197.59	4,605.81
3	4,605.81	221.60	23.03	198.57	4,407.24
4	4,407.24	221.60	22.04	199.57	4,207.67
5	4,207.67	221.60	21.04	200.56	4,007.11
6	4,007.11	221.60	20.04	201.57	3,805.54
7	3,805.54	221.60	19.03	202.58	3,602.96
8	3,602.96	221.60	18.01	203.59	3,399.37
9	3,399.37	221.60	17.00	204.61	3,194.77
10	3,194.77	221.60	15.97	205.63	2,989.14
11	2,989.14	221.60	14.95	206.66	2,782.48
12	2,782.48	221.60	13.91	207.69	2,574.79
13	2,574.79	221.60	12.87	208.73	2,366.06
14	2,366.06	221.60	11.83	209.77	2,156.29
15	2,156.29	221.60	10.78	210.82	1,945.47
16	1,945.47	221.60	9.73	211.88	1,733.59
17	1,733.59	221.60	8.67	212.94	1,520.66
18	1,520.66	221.60	7.60	214.00	1,306.66
19	1,306.66	221.60	6.53	215.07	1,091.59
20	1,091.59	221.60	5.46	216.15	875.44
21	875.44	221.60	4.38	217.23	658.22
22	658.22	221.60	3.29	218.31	439.90
23	439.90	221.60	2.20	219.40	220.50
24	220.50	221.60	1.10	220.50	0.00

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Step 3: Interest and principal over time

- The sum of interest and principal reduction always equals the payment
- The interest payments fall at an increasing rate as principal reductions reduce future interest charges



A quick way to compute total interest paid:
Total interest = Payment × Months − Loan amount

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The MBE and Loan Amortization

- From the bank's point of view, the loan is an asset
  - For the borrower, it's a liability
- At inception, the asset is worth $V_0 = \$5000$.
  - What is it worth at the end, i.e. what is V_1?
- The MBE suggests (since r includes the risk premium):
  - $rV_0 = (V_1 - V_0) + C$
  - Monthly Interest = (End balance – Begin balance) + Payment, or
  - End balance = Begin balance + Monthly Interest – Payment
    - Also, Payment – Monthly Interest = Principal Reduction



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The bank's analysis of your loan

- Assumptions
  - The bank's cost of capital is 1.2% per year (monthly rate 0.1%)
  - Borrowers in this loan cohort have an average default rate of 20 bppm (basis points per month)
  - On default in any given month, no further payments are received
- Analysis
  - The bank has been promised 24 payments of 221.60
  - The probability of default in month 1 is 20 bp, and no payments are ever received in this case
  - The probability of default in month 2 is 20 bp of 99.8%, or (0.002×0.998), since a borrower has to survive month 1 to default in month 2
  - The present value of the loan is the present value of the payments in each default month, taken at the bank's cost of capital.
  - The overall loan value is the probability-weighted PV of the possible default outcomes

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The bank's credit spreadsheet

Present value of payments received by default month

Payments received given default month

Probability of default in given month

Probability-weighted PV = Expected PV = $5,123.73

The no-default case

Month	Prob	Payments received	PV receipts
1	0.2000%		0
2	0.1996%	1	$221.38
3	0.1992%	1 1	$442.54
4	0.1988%	1 1 1	$663.48
5	0.1984%	1 1 1 1	$884.20
6	0.1980%	1 1 1 1 1	$1,104.70
7	0.1976%	1 1 1 1 1 1	$1,324.98
8	0.1972%	1 1 1 1 1 1 1	$1,545.04
9	0.1968%	1 1 1 1 1 1 1 1	$1,764.87
10	0.1964%	1 1 1 1 1 1 1 1 1	$1,984.49
11	0.1960%	1 1 1 1 1 1 1 1 1 1	$2,203.89
12	0.1956%	1 1 1 1 1 1 1 1 1 1 1	$2,423.07
13	0.1953%	1 1 1 1 1 1 1 1 1 1 1 1	$2,642.03
14	0.1949%	1 1 1 1 1 1 1 1 1 1 1 1 1	$2,860.77
15	0.1945%	1 1 1 1 1 1 1 1 1 1 1 1 1 1	$3,079.30
16	0.1941%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$3,297.60
17	0.1937%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$3,515.69
18	0.1933%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$3,733.56
19	0.1929%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$3,951.21
20	0.1925%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$4,168.65
21	0.1922%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$4,385.86
22	0.1918%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$4,602.86
23	0.1914%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$4,819.65
24	0.1910%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$5,036.21
25	95.3088%	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	$5,252.56

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The bank's business analysis

- The profit on this loan is $123.73 in present value terms
- The credit spread is the difference between the YTM (based on the promised payments) and the bank's cost of capital, i.e. 20 bppm.
  - Monthly YTM is the implied rate of return earned on the annuity valued at 5123.73
  - Annual YTM = Monthly YTM × 12 (no compounding)
  - This (no compounding) standard applies to all loans and bonds
- Is it a coincidence that the credit spread approximately equals the default rate?

	Monthly	Annual
Bank's cost of capital	0.10%	1.20%
Default rate	0.20%	2.40%
Expected loan value	$5,123.73	
Yield to maturity (promised)	0.300601%	3.61%
Credit spread	0.200601%	2.41%
Loan amount	$5,000.00	
Net present value	$123.73	

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Oh no, not the MBE again!

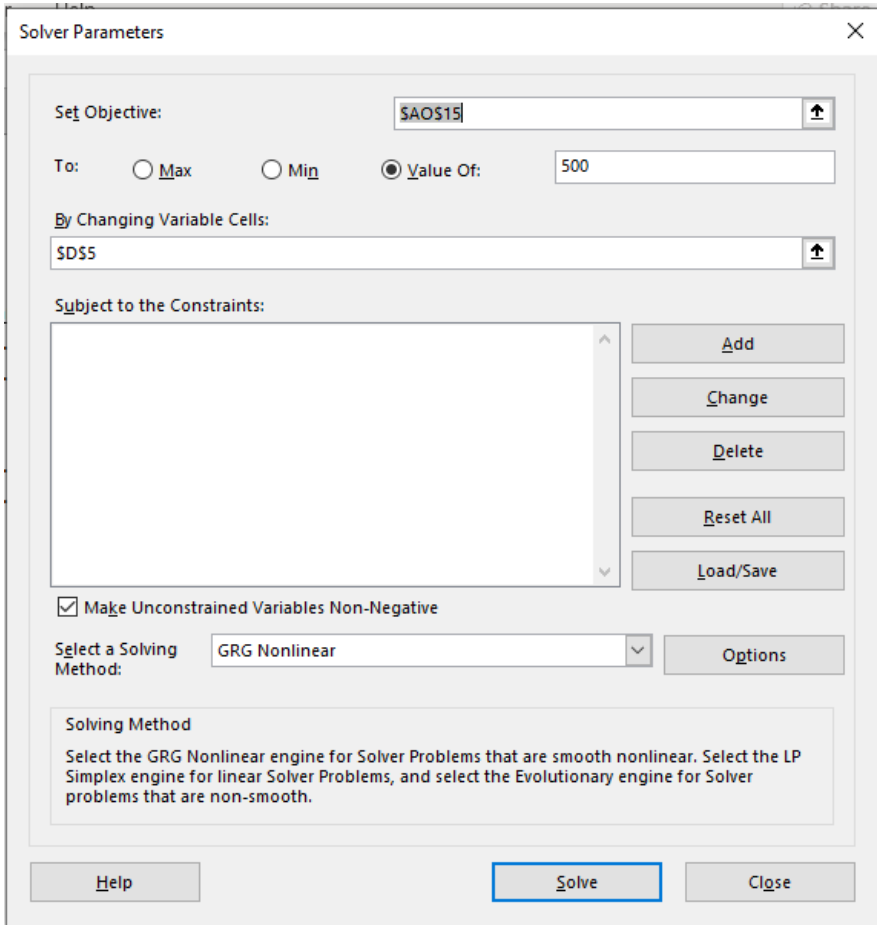
- The expected cash flows are declining at a rate of 20 bppm
- Use the growing annuity formula
  - g = -0.002
  - r* = (0.001 + 0.002)/(1 – 0.002) = 0.00300601
- The YTM exactly equals r*
  - The credit spread is $r^* - r = \pi(1 + r)/(1 - \pi) \equiv p \approx \pi$
- The time 0 payment is the mortgage payment amount of $221.60.
- The loan value is $5123.73 (exactly the same)
- Since the expected loss every month is 0.002 times the loan value, the charge is proportional
- In this special case, the MBE includes an expected default loss as part of the capital gain:
- $rV_0 = \left(-\frac{C}{(1+r)^n} - pV_0\right) + C \quad \longrightarrow \quad r^* V_0 = -\frac{C}{(1+r)^n} + C$



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Bonus Excel Question

- As the bank manager, you've been directed to make the profit on each loan equal to 10% of loan value
- With all other assumptions the same, what would the contracted loan rate have to equal?
- HINT: Use Excel Solver
  - Install the "Solver" add-in in Excel
  - Also install the Data Analysis Toolpak for statistical analysis, you'll need it later
  - Call these from your DATA tab
- Use Excel to find the value of the contract rate that makes the profit $500 for the $5000 loan
- Answer: 13.07%



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Further complexities in loan modeling

- Consumers often prepay their loans
- Default rates are not constant and are not easy to predict
  - Machine learning models are becoming more popular in retail credit underwriting
- Recoveries on some loans are possible
- Shifts in macroeconomic variables cause changes in fraud rates, default rates, prepayments and recoveries (think COVID)
- This challenging space provides job opportunities for aspiring financial engineers

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Discussion Question (time permitting)

- In what sense does a bank "value" its customers?
- E.g.
  - High credit ratings?
  - History of missed loan payments?
  - Savings rate?
  - Risk aversion?
  - Age?
- Example: Chase vs. Capital One

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Summary

- Excel provides an easy paradigm and a popular standard for performing present value calculations and solving for missing parameters
- Excel also provides What-if analysis, graphics, and goal-seeking capabilities (among other things we shall discover)
- Consumer loans provide not only a present value application, but a chance to do more advanced financial analysis and business decision-making, such as credit underwriting
- Every student of finance must understand amortization tables and their uses
- These simple analyses form the basis for more advanced problem solving, such as default assessment and credit pricing.

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Review Questions for Recitation

1. What are the 5 parameters of the standard Excel present value paradigm?
2. What parameter is not included in Excel?
3. How does one compute an amortization table?
4. What are two ways to find a loan balance after a given number of months?
5. What is the easiest way to compute the total interest paid over the life of the loan?
6. How does one incorporate default probabilities into the analysis of a loan amortization?
7. If a loan has a constant default probability per period π, and the risk-free rate is r, what is the loan's yield to maturity? What is its credit spread?
8. Name three factors that make analysis of retail loans difficult.

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Loan terminology

- Contract rate
- Amortize/amortization
- Principal
- Interest
- Payment
- Principal reduction
- Asset
- Liability
- Cost of capital
- bppm
- Cohort
- Underwriting
- Default rate
- Prepayment
- YTM
- Loss given default (LGD)
- Recovery

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Key formulas

- Amortization table formulas (monthly loans)
  - Payment = PMT(Rate/12,nper,loan amt)
  - Interest = Begin × Rate/12
  - Principal reduction = Payment – Interest
  - End = Begin – Principal reduction, or
  - End = Begin + Interest - Payment
  - Total interest paid = Payment × # of payments – Loan amount
- IRR on a loan with constant conditional full default rates (no recovery) and no additional risk premium
  - $r^* = \frac{r+\pi}{1-\pi}$
- Credit spread on that loan
  - $CS = r^* - r$

Financial calculation	Excel syntax
Present value of annuity including a final payment of FV	=pv(rate, nper, pmt, [fv], [type])
Payment that makes present value equal to target value	=pmt(rate, nper, pv, [fv], [type])
Number of payments needed to match present value to target value	=nper(rate, pmt, pv, [fv], [type])
Interest rate per period that matches present value and target value	=rate(nper, pmt, pv, [fv], [type])
Final payment that matches present value and target value	=fv(rate, nper, pmt, [fv], [type])
Present value of a range of cash flows starting in one period	=npv(rate, range of values)
Rate of return that makes the present value of a stream of payments zero (the first payment is immediate)	=irr(range of values, [guess])

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